BIRMINGHAM CITY University

CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Project Interim Report

The application for learning Chinese and the other languages

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1.0 Introduction and Context

According to development of artificial intelligence and internet, more people have more choices for learning the other languages. In this background, Automatic Speech Recognition (ASR) and computer vision provide new possibilities for computer assisted learning (CAL).

Nowadays, most languages-learning applications are focused on vocabulary memory or the grammar correction. However, there is not very detailed feedback on pronunciation of a word in different sentences. This would direct influence the confidence of users and the understanding of sentences.

In research on Chinese as a second language learning, mastering tones has long been considered one of the most crucial elements of Chinese language learning. Learners can often understand vocabulary and grammatical structures relatively quickly, but they struggle to accurately grasp the pitch patterns and corresponding meanings of the four tones. This difference not only affects the ease of communication but also limits the practical effectiveness of automated speech learning systems.

There is a search shown that the incorporation of visual information significantly enhances the perceptual efficacy of Mandarin tones, particularly demonstrating pronounced benefits for novice learners. (Han et al., 2019; Garg et al., 2023)

Based on the above, this project looks for a new method for learning method on mobile which mixed ASR and CAL. By using real-time and different styles feedback interactive experience to improve the efficiency and immersion of learning Chinese tones.

in recent years, with the improvement of globe development and communication, the number of Chinese learners who are not native speaker is increased. According to data from World Language Learning Trends Report (2023), there are more than 30 million leaners are learning on mobile devices. However, most learners are still on grammar, words and listening parts but miss the learning of tone and pronunciation part. At the same time, Artificial intelligence (AI) educational technology is rapidly penetrating the field of language learning. From speech recognition to adaptive learning algorithms. AI is redefining the boundaries of "personalized learning". Against this backdrop, this project is not only a technological innovation for teaching Chinese tones, but also an exploration of the "human-computer interactive language learning model". The ultimate goal of the project is to establish an intelligent system that can capture learners' pronunciation movements in real

time, provide multi-dimensional feedback, and achieve dynamic self-correction during the learning process.

1.1 Problem Statement

Currently, most Mandarin learning applications still have some problems. For example, most system only show correct or wrong feedback. In this case, users do not know where is wrong and how to figure out this. Moreover, these applications miss the lack of visual pronunciation cues such as lip movements and jaw movements. That will make users hard to learn correct movement for correct pronunciation. Therefore, beginners are hard to get the targeted feedback when practicing independently. Furthermore, the survey found that over 70% of Chinese beginners share common problems during the self-study stage of pronunciation, namely, "not being able to hear mistakes" and "not being able to correct the problems." This indicates that current learning tools on the market still have significant room for improvement in terms of feedback mechanisms and interactive experiences.

Existing research and engineering practices have provided that facial key point tracking technology, Like MediaPipe Face Landmark and Face mesh have already run stable on mobile devices, and this provide the way to achieve the visualized pronunciation feedback (Google, 2024).

2.0 Review of Existing Knowledge

Research from Han at et., 2019 shows that under the same hardware conditions, when tested separately for auditory and visual perception only, learner’s tone recognition accuracy was significantly higher than that for auditory only. Also, In a systematic review of various types of language learning, (Zhang and Yuan,2024) pointed out that integrating auditory and visual input can enhance learners' perception and memory of tones, which is a future trend in second language phonetics teaching.

At the technical level, (He et al.,2019) proved that the streaming end-to-end speech recognition model based on RNN-T can achieve real-time speech recognition on mobile devices, while (Jacob et al., 2018) verified that 8-bit integer quantization technology can significantly reduce power consumption and latency while maintaining accuracy.

Recent interdisciplinary research in speech therapy and second language teaching has shown that learners can develop spatial concepts and muscle memory for

pronunciation more quickly when observing images of their own pronunciation (Bliss et al., 2018). This directly utilizes learners' own learning habits, replacing abstract descriptions with more realistic images, and enabling them to more efficiently identify their errors. This provides psychological support for the design of this study and leverages the brain itself.

Totally, these achievements lay the theoretical and engineering foundation for developing a mobile learning system that integrates auditory and visual feedback.

2.1 Critical Analysis and Gap Identification

While some research has yielded certain results, there are still areas that have not been studied or have not yet been investigated. For example, the granularity of tone feedback is insufficient; most systems can only judge right or wrong, lacking analysis of error type and direction (Han et al., 2019). Furthermore, the integration of visual guidance is not high. Although mobile facial tracking technology is mature, it has not yet been applied to pronunciation training systems (Garg et al., 2023). In addition, over-reliance on cloud computing can lead to risks related to latency, energy consumption, and the preservation of user privacy (He et al., 2019). Furthermore, there is limited data and research available for beginners. Most current research focuses on learners with some prior knowledge or native speakers, without directly addressing users with specific needs or real-world teaching scenarios. In these cases, the timeliness and comprehensibility of feedback are more crucial to learning outcomes than mere recognition accuracy.

These gaps are precisely where the project finds its foothold. The market needs a beginner-centric learning system that guarantees speed, effectiveness, and visibility, thereby reducing the learning costs and increasing the learning efficiency for beginners.

From the other side, this project still can be improved more. Till now, the project only focuses on learners' tone. This is very poor functional as a learning application.

2.2 Project Justification

This project aims to do some improve based on the aforementioned research gaps by proposing solutions. By using the local acoustic analysis and keypoint-based visual feedback on mobile devices, the system can achieve real-time tone recognition and lip-reading guidance while ensuring privacy (Google, 2024; Jacob et al., 2018).

Studies by (Han et al., 2019) and (Garg et al., 2023) provide ample evidence that audiovisual training methods can effectively improve learners' tone recognition and output performance.

Furthermore, this project adopts a modular design, allowing for future expansion beyond Mandarin to other tonal languages (such as Vietnamese and Thai) and clinical fields such as speech rehabilitation and pronunciation correction, demonstrating significant academic and applied value.

For showing more research significance, the project will import the multiple feedback system which depends on the tone of learners speak. The system will show different colours with the varying degrees of errors. For example, when the learner's tone deviates little from the target standard tone, the system will use a bright colour to indicate this or provide a voice prompt to inform the user that they are "close to correct." Conversely, when the learner's tone deviates significantly from the target standard tone, a darker colour will be used to indicate this. Therefore, this system can create a more motivating learning experience. This method references the findings of (Olson,2021), which demonstrates that refined feedback can significantly improve the efficiency of second language pronunciation correction. Besides, the project will set two groups to verify the adaptation of the system. The first group is controlling experimental group. The second group is delaying test group. For first group, the project wants to test the effects for learning. For second group, this group will gap one week time between the first learning and the second learning which for testing out the retention rate of learner’s tone during gap time. This test design references the analysis of the persistent learning effect of visual feedback by (Bliss et al., 2018), thus ensuring that the research results are more universal and scientific.

In terms of data analysis, the research will employ a comprehensive evaluation model combining quantitative and qualitative methods. The quantitative component will primarily use metrics such as Mean Absolute Error (MAE) and Accuracy of Recognition (ACC) to measure system performance; the qualitative part will utilize questionnaires and usage feedback to understand learners' subjective experiences. All experimental data will be anonymized and stored locally using encrypted methods. Finally, this study is expected to not only provide a new breakthrough for learning Mandarin tones, but also to provide a reusable technical framework for multilingual speech teaching and rehabilitation training systems, laying some foundation for future interdisciplinary educational applications.

3.0 Project Aims, Objectives, and Scope

3.1 Project Aim

This project aims to research and develop a smart mobile Chinese learning application called "Visible Spoken Language."

The system will integrate local speech recognition and facial landmark tracking to achieve real-time, fine-grained tone assessment and visual feedback. It will focus on solving problems such as feedback latency, unclear tones, and lack of pronunciation guidance, providing learners with an efficient, low-latency, and privacy-conscious learning experience.

In addition, if time permits after achieving the above goals, this project will also add a learning module on trending internet slang, enabling users to understand the meaning and pronunciation of some internet terms. This will increase the app's usage rate.

3.2 Project Objectives

To achieve the above objectives, this project tentatively outlines the following six phased tasks:

1. Literature and Product Research (October–November 2025): Systematically review mobile speech recognition, tone recognition, and face tracking technologies; read relevant literature and identify feasible solutions and performance standards. If possible, develop relevant conceptual software first and then optimize that.
2. Speech and tone recognition module prototype (November–December 2025): Build a streaming local speech recognition model and integrate tone classification, with the goal of achieving low-latency response on mid-range Android devices.
3. Facial Key Point Tracking and Visualization (December 2025 – January 2026): Enables real-time detection of lips and jaw and generates a comparative pronunciation visualization interface.
4. System Integration and Interface Design (January-February 2026): Integrate the voice and visual modules into a complete prototype and develop an intuitive and highly interactive user interface.
5. User Testing and Learning Outcome Evaluation (March 2026): Recruit 12–20 beginners to conduct tests and evaluate tone accuracy, learning time, and

subjective satisfaction of the application. Also the difficulty of using the application.

6. Data Analysis and Report Writing (April 2026): Conduct a comprehensive analysis of quantitative and qualitative data and summarize system performance and teaching implications.

3.3 Project Scope

The research scope includes:

- Platform: Android smartphones and tablets.
- Corpus: Chinese Pinyin syllables with four tones.
- Visual tracking: Real-time lip and jaw movement detection.
- Target users: Beginner to intermediate Chinese learners.
- Evaluation metrics: Tone recognition accuracy, processing latency, learner satisfaction, and pronunciation improvement.

Excluded from the scope of this study:

- iOS or cross-platform deployment.
- Advanced grammar and semantic learning.
- Continuous speech recognition tasks.
- Teacher-side data analysis and cloud-based visualization services.

4.0 Project Design

This study employs a user-centred design methodology, combining quantitative and qualitative analysis to ensure the scientific rigor and effectiveness of the system development for instruction.

The system design process comprises four phases: requirements analysis, prototype development, functional verification, and instructional experimentation. The prototype development phase will utilize an agile methodology, iteratively updating and optimizing the interface and algorithm logic to guarantee the system's usability and stability.

In the experimental phase, data will be collected through questionnaires and user tests, and interactive experience analysis will be conducted in conjunction with learner feedback. This hybrid research approach, combining technology and questionnaires, allows for a more comprehensive evaluation of the application's instructional efficiency and value.

4.1 Design and Methods

This project employs a combination of development-based research and experimental validation. The research objective is to build and validate a learning system capable of achieving real-time tone recognition and visual feedback on mobile devices.

The entire research process was divided into three main stages:

1. System Development Stage: A lightweight tone recognition model was built based on Python and TensorFlow Lite, and the MediaPipe framework was used to implement key point detection of the lips and jaw. The speech module and vision module were integrated and tested in an Android environment.
2. Usability testing phase: Invite 12–20 beginners to participate in the usage experiment and collect data on their usage time, tone accuracy, and subjective experience.
3. Effectiveness Verification Phase: By comparing the results of individual tests with those of subsequent small-scale public tests, the system's role in promoting learning effectiveness is evaluated, and qualitative analysis is conducted using questionnaires and interviews.

Existing research has shown that adopting a user-centred design approach can significantly improve the usability of learning applications (Han et al., 2019). Therefore, in the initial stages of development, this project will use small-sample user interviews or anonymous questionnaires to understand learners' actual needs for pronunciation feedback and visual guidance, thereby optimizing the user interface and prompting mechanisms.

4.2 Justification of Methods

The main reasons for choosing this research method are as follows:

High reproducibility: Developed based on open-source frameworks (TensorFlow Lite, MediaPipe), it is easy to promote in academic and teaching settings.

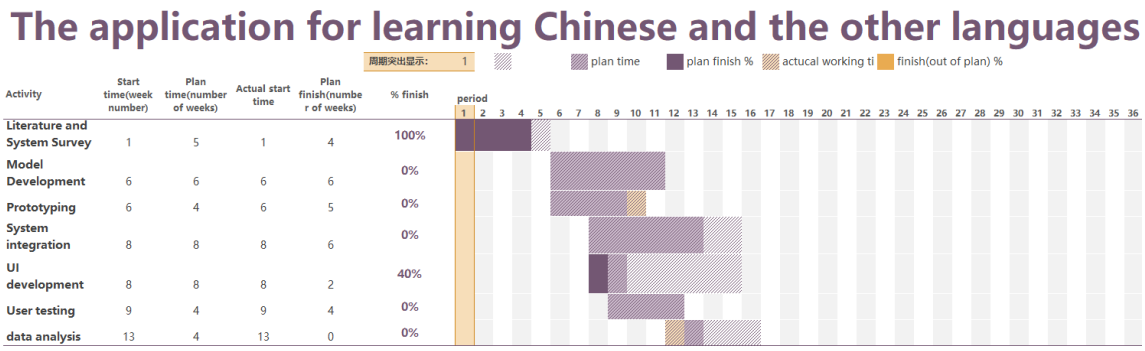
Low latency: On-device inference avoids network transmission latency, suitable for real-time feedback requirements.

Strong security and privacy: Data processing is completed locally, without cloud uploads, reducing learner privacy risks.

Good scalability: The modular design of the system facilitates future expansion to other language and speech training scenarios.

Studies by Jacob et al., 2018 and He et al., 2019 have shown that quantized networks and end-to-end streaming models can effectively reduce latency and energy consumption, making real-time speech recognition on mobile devices possible. Therefore, this project adopts an integer quantization strategy in model optimization to ensure smooth operation on mobile devices.

4.3 Project Timeline



Activity	Start time(week number)	Plan time(number of weeks)	Actual start time	Plan finish(number of weeks)	% finish
Literature and System Survey	1	5	1	4	100%
Model Development	6	6	6	6	0%
Prototyping	6	4	6	5	0%
System integration	8	8	8	6	0%
UI development	8	8	8	2	40%
User testing	9	4	9	4	0%
data analysis	13	4	13	0	0%

Figure 1: Gantt chart for plan of this project.

5.0 Feasibility, Risks, and Ethical Aspects

5.1 Feasibility

The project's feasibility is primarily reflected in the following resource and technical aspects:

Lightweight integration of speech and vision modules can be achieved using frameworks such as TensorFlow Lite and PyTorch Mobile, combined with Android Studio.

Existing research (He et al., 2019; Google, 2024) has demonstrated the feasibility of real-time speech recognition and key point detection on mid-range mobile devices.

Furthermore, our research team possesses the necessary programming and educational backgrounds, enabling them to independently complete system development and teaching experiments.

Both hardware and software costs can be kept within the range of student-level research projects, the project timeline is reasonable, and it demonstrates good operability and expected output.

5.2 Risk Analysis and Mitigation

Potential Risk	**Likelihood (Low/Med/High)**	**Impact (Low/Med/High)**	**Mitigation Strategy**
Model latency higher than expected	*Medium*	*Medium*	*Use smaller model and use GPU to increase the speed of dealing.* *Optimize the length of input.*
Not enough testers	*Low*	*Medium*	*Recruiting testers in advance.* *Increase the methods for testing results.*
Model accuracy *Not correct* *enough*	*Medium*	*High*	*Extend more language database.* *Employing transfer learning and data augmentation*
System compatibility issues	*Low*	*Medium*	*Do cross-testing on different platforms.*

Table 1: The table for risk analysis and solutions.

Ps: To make sure the development period walks well. The project sets the check point in each two weeks. In this case, the project can find out the problems and fix problems as soon as possible at the check point time.

5.3 Ethical Considerations

The research strictly adheres to academic ethical standards to ensure the security and privacy of participants' data. All voice and visual data are processed locally on the device and are not uploaded to any cloud server.

Before the experiment, researchers will explain the experiment's content, risks, and rights to participants both in writing and orally to ensure they fully understand the experiment and participate voluntarily. Any image data and voice data collected during the study will be blurred to prevent identity verification. Moreover, the project will delete all data from the test when the project before end to make sure ethical compliance.

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Appendices